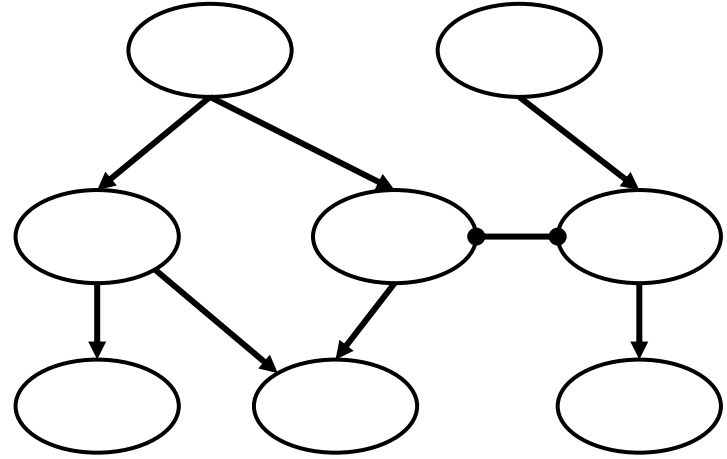




Lecture 8 (Routing 5)

Inter-Domain Routing Exercises ANS

Q1. BGP

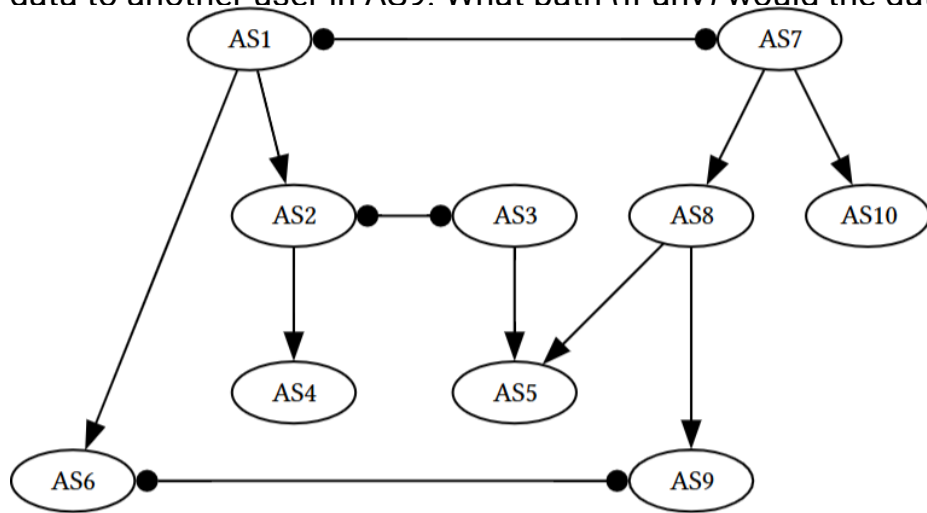
Consider the following configuration of autonomous systems and their relationships.

A square-triangle line indicates a provider-customer relationship, and a circle-circle line indicates a peer relationship. We have two tier 1 providers (AS1 and AS7). Assume we are using the following selection and export rules for all ASes:

Selection: Prefer to send data through customers > peers > providers in order to minimize costs.

Export: To customers, advertise all paths. To all other ASes, advertise only paths to customers (and paths to your own AS, of course).

- 1) A user in AS4 tries to send data to a friend in AS5. What path (if any) would the data take to get to the friend?
- 2) A user in AS6 tries to send data to a friend in AS5. What path (if any) would the data take to get to the friend?
- 3) A user in AS1 tries to send data to another user in AS9. What path (if any) would the data take?



Q. BGP ANS

1) A user in AS4 tries to send data to a friend in AS5. What path (if any) would the data take to get to the friend?

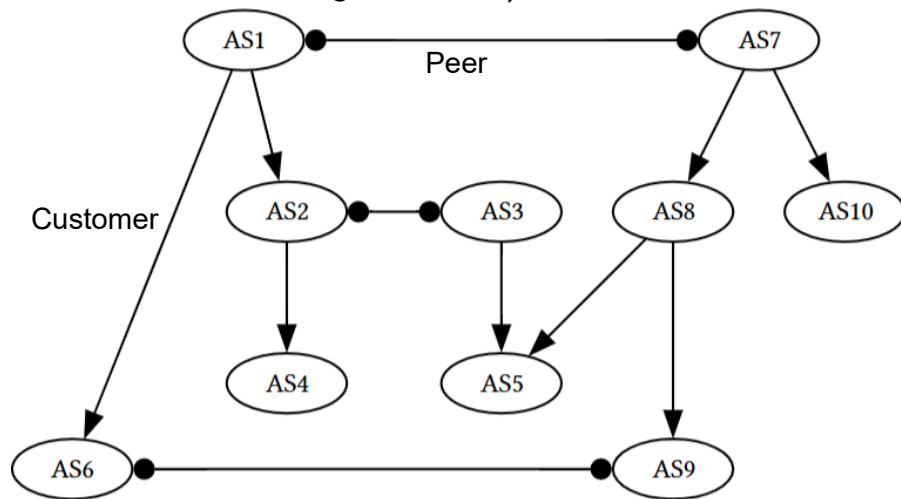
ANS: $AS4 \rightarrow AS2 \rightarrow AS3 \rightarrow AS5$

2) A user in AS6 tries to send data to a friend in AS5. What path (if any) would the data take to get to the friend?

ANS: $AS6 \rightarrow AS1 \rightarrow AS7 \rightarrow AS8 \rightarrow AS5$. Since AS2 won't advertise its peer's (AS3's) path (about how to reach AS5) to its provider (AS1), so AS1 doesn't know it can reach AS5 through AS2.

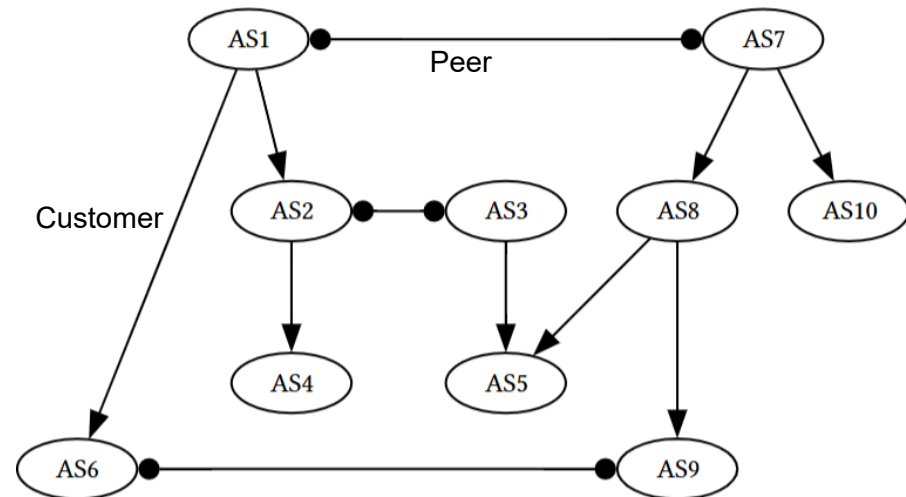
3) A user in AS1 tries to send data to another user in AS9. What path (if any) would the data take?

ANS: $AS1 \rightarrow AS7 \rightarrow AS8 \rightarrow AS9$. Same reasoning as Part 2).



Explanations

- 1) $AS4 \rightarrow AS2 \rightarrow AS3 \rightarrow AS5$. This path respects all Gao-Rexford rules:
 - $AS4$ has a customer neighbor ($AS2$ will accept)
 - $AS2$ has a peer neighbor ($AS3$) AND a customer neighbor ($AS5$ paying $AS3$)
 - $AS3$ has a customer neighbor ($AS5$ paying)
 - Path is valley-free (no going up after going down)
- 2) $AS6 \rightarrow AS1 \rightarrow AS7 \rightarrow AS8 \rightarrow AS5$. Since $AS2$ won't advertise its peer's ($AS3$'s) path (about how to reach $AS5$) to its provider ($AS1$), so $AS1$ doesn't know it can reach $AS5$ through $AS2$.
 - Export policy: $AS2$ receives a route from peer $AS3$ (about reaching $AS5$)
 - Export rule: Routes from peers are only advertised to customers (not to providers)
 - Therefore: $AS1$ ($AS2$'s provider) does NOT learn about the $AS3 \rightarrow AS5$ path
 - So $AS1$ can only reach $AS5$ via $AS7$ (its peer) and then $AS8 \rightarrow AS5$
 - This forces the long path through Tier 1 providers



Explanations

- 3) $AS1 \rightarrow AS7 \rightarrow AS8 \rightarrow AS9$. This path respects all Gao-Rexford rules:
 - AS1 is a Tier 1 AS (no providers)
 - AS7 is also a Tier 1 AS (peering with AS1)
 - To reach AS9, AS1 needs a path advertised by AS7
 - AS7 receives routes from customers (AS8) and peers (AS1)
 - Export rule: AS7 advertises customer routes (AS8's path) to all ASes
 - So AS1 can use: $AS1 \rightarrow AS7 \rightarrow AS8 \rightarrow AS9$
- The principle is the same (export policies hide certain paths), but the mechanics are slightly different. Comparison:
 - Part 2: AS1 tries to reach AS5 through AS2 (but AS2 hides the peer path)
 - Part 3: AS1 tries to reach AS9 through AS7 (which advertises its customer's route)

